

Safety data sheet

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BASF Safety data sheet
Date / Revised: 08.12.2025
Product: **MELFLUX 5581 F MB**

Version: 2.0

(30857638/SDS_GEN_SG/EN)

Date of print: 09.12.2025

1. Substance/preparation and manufacturer/supplier identification

Product name:
MELFLUX 5581 F MB

Use: Product for construction chemicals

Manufacturer/supplier:
BASF South East Asia Pte Ltd.
128 Beach Road #18-01
Guoco Midtown, 189773, Singapore
Telephone: +65 8322 4420
Telefax number: +65 6 334-0330
E-mail address: benny.zou@basf.com

Emergency information:
Singapore Emergency Toll-Free Number:
Telephone: 1800-723-1361
International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification of the substance and mixture:
No need for classification according to GHS criteria for this product.

Label elements and precautionary statement:

The product does not require a hazard warning label in accordance with GHS criteria.

Other hazards which do not result in classification:
The product is under certain conditions capable of dust explosion.
May form explosible dust-air mixture if dispersed

3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

Polymer based on: polycarboxylate ether

No particular hazards known.

4. First-Aid Measures

General advice:

| Remove contaminated clothing.

If inhaled:

| If difficulties occur after dust has been inhaled, remove to fresh air and seek medical attention.

On skin contact:

| Wash thoroughly with soap and water

On contact with eyes:

| Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:

| Rinse mouth and then drink 200-300 ml of water. Do not induce vomiting unless told to by a poison control center or doctor.

Note to physician:

| Symptoms: (Further) symptoms and / or effects are not known so far

| Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:

| dry powder, foam

Unsuitable extinguishing media for safety reasons:

| carbon dioxide

Additional information:

| Avoid whirling up the material/product because of the danger of dust explosion.

Specific hazards:

| harmful vapours

| Generation of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:

| Wear a self-contained breathing apparatus.

Further information:

The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures

Personal precautions:

Avoid dust formation. Use personal protective clothing.

Environmental precautions:

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:

For small amounts: Pick up with suitable appliance and dispose of.
For large amounts: Contain with dust binding material and dispose of.
Avoid raising dust.

7. Handling and Storage

Handling

Breathing must be protected when large quantities are decanted without local exhaust ventilation.

Protection against fire and explosion:

Avoid dust formation. Take precautionary measures against static discharges.

Storage

Further information on storage conditions: Keep only in the original container in a cool, dry, well-ventilated place away from ignition sources, heat or flame. Protect from direct sunlight. The substance/product may cake at higher temperatures/pressure.

Protect from temperatures above: 40 °C

The packed product must be protected against exceeding the indicated temperature.

8. Exposure controls and personal protection

Components with occupational exposure limits

Silicon dioxide, 7631-86-9;

TWA value 10 mg/m³ (OEL (SG))

TWA value 10 mg/m³ (ACGIHTLV), Inhalable particles

TWA value 3 mg/m³ (ACGIHTLV), Respirable particles

TWA value 10 mg/m³ (OEL (SG)), Particulate

Personal protective equipment

Respiratory protection:

Suitable respiratory protection for lower concentrations or short-term effect: Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:

Chemical resistant protective gloves

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):

e.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), polyvinylchloride (0.7 mm) and other

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Safety glasses with side-shields.

Body protection:

light protective clothing

General safety and hygiene measures:

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is recommended. When using, do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift. At the end of the shift the skin should be cleaned and skin-care agents applied. Gloves must be inspected regularly and prior to each use. Replace if necessary (e.g. pinhole leaks).

9. Physical and Chemical Properties

Form: powder
Colour: Yellowish to brownish
Odour: characteristic
Odour threshold: No applicable information available.

pH value: approx. 6.5 - 8.5
(20 %(m), 20 °C)

Melting temperature: The substance / product decomposes therefore not determined.

boiling temperature: not applicable

Flash point: not applicable

Evaporation rate: The product is a non-volatile solid.

Flammability (solid/gas): No data available.

Lower explosion limit: 30 g/m³

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Upper explosion limit:	No data available.	
Ignition temperature:	360 °C	(BS EN 50281-2-1 Method B)
Thermal decomposition:	> 180 °C No decomposition if stored and handled as prescribed/indicated.	
Self ignition:	Temperature: 165 °C Data for powdery solid. No self ignition was observed up to the specified temperature.	Test type: Self-ignition at high temperatures. (Method: VDI 2263, sheet 1, 1.4.2 (May 1990))
Self heating ability:	It is not a substance capable of spontaneous heating according to UN transport regulations class 4.2.	(UN Test N.4 (self heating substances))
Minimum ignition energy:	30 - 100 mJ Inductivity: 1 mH Grain size distribution: < 63 µm	
Explosion hazard:	not explosive	
Fire promoting properties:	not fire-propagating	
Vapour pressure:	The product has not been tested.	
Density:	not applicable	
Relative density:	No data available.	
Bulk density:	approx. 300 - 550 kg/m ³	
Relative vapour density (air):	The product is a non-volatile solid.	
Solubility in water:	soluble (20 °C)	
Miscibility with water:	not applicable	
Hygroscopy:	The product has not been tested.	
Partitioning coefficient n-octanol/water (log Pow):	not applicable for mixtures	
Viscosity, dynamic:	not applicable, the product is a solid	
Viscosity, kinematic:	No data available.	

Other Information:

If necessary, information on other physical and chemical parameters is indicated in this section.

Particle characteristics

Particle size distribution:	< 300 µm	(D50, Volumetric Distribution, ISO 13322-2)
Specific Surface Area:	No data available.	
Particle Shape:		

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Dustiness: No data available.
No data available.

10. Stability and Reactivity

Conditions to avoid:
See SDS section 7 - Handling and storage.

Thermal decomposition: > 180 °C
No decomposition if stored and handled as prescribed/indicated.

Substances to avoid:
strong acids, strong bases, strong oxidizing agents, strong reducing agents

Corrosion to metals: Corrosive effects to metal are not anticipated.

Hazardous reactions:
Dust explosion hazard.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:
The product is stable if stored and handled as prescribed/indicated.

Reactivity:
No hazardous reactions if stored and handled as prescribed/indicated.

11. Toxicological Information

Routes of exposure

Acute oral toxicity

Experimental/calculated data:
LD50rat (oral): > 5,000 mg/kg
The product has not been tested. The statement has been derived from the properties of the individual components.

Acute inhalation toxicity

LC50 rat (by inhalation): 4 h
not determined

Acute dermal toxicity

LD50 rat (dermal):
not determined

Assessment of acute toxicity

Virtually nontoxic after a single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact. Based on available data, the classification criteria are not met.

Symptoms

| (Further) symptoms and / or effects are not known so far

Irritation

Assessment of irritating effects:

No irritation is expected under intended use and appropriate handling. Based on available data, the classification criteria are not met.

Experimental/calculated data:

Skin corrosion/irritation rabbit: non-irritant

The product has not been tested. The statement has been derived from the properties of the individual components.

Serious eye damage/irritation rabbit: non-irritant

The product has not been tested. The statement has been derived from the properties of the individual components.

Respiratory/Skin sensitization

Assessment of sensitization:

Based on the ingredients, there is no suspicion of a skin-sensitizing potential. Based on available data, the classification criteria are not met.

Experimental/calculated data:

guinea pig: Non-sensitizing.

Germ cell mutagenicity

Assessment of mutagenicity:

| Based on the ingredients, there is no suspicion of a mutagenic effect.

Carcinogenicity

Assessment of carcinogenicity:

| The whole of the information assessable provides no indication of a carcinogenic effect.

Experimental/calculated data:

No data available.

Reproductive toxicity

Assessment of reproduction toxicity:

| Based on the ingredients, there is no suspicion of a toxic effect on reproduction.

Experimental/calculated data:

No data available.

Developmental toxicity

Assessment of teratogenicity:

| Based on the ingredients, there is no suspicion of a teratogenic effect.

Specific target organ toxicity (single exposure)

Based on available data, the classification criteria are not met.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

| The information available on the product provides no indication of toxicity on target organs after repeated exposure.

Experimental/calculated data:
No data available.

Aspiration hazard

| not applicable

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:

Based on available data, the classification criteria are not met. There is a high probability that the product is not acutely harmful to aquatic organisms.

Toxicity to fish:

| LC50 (96 h), Fish (other)
| not determined

Aquatic invertebrates:

| LC50 (48 h), daphnia (other)
| not determined

Aquatic plants:

| EC50 (72 h), algae (other)
| not determined

Microorganisms/Effect on activated sludge:

| EC50 (0.5 h), bacteria (other)
| not determined

Chronic toxicity to fish:

| No data available.

Chronic toxicity to aquatic invertebrates:

| No data available.

Assessment of terrestrial toxicity:

| No data available concerning terrestrial toxicity.

Mobility

Assessment transport between environmental compartments:
No data available.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):
Not readily biodegradable (by OECD criteria).

Bioaccumulation potential

Assessment bioaccumulation potential:
No data available.

Additional information

Add. remarks environm. fate & pathway:
Treatment in biological waste water treatment plants has to be performed according to local and administrative regulations.

Other ecotoxicological advice:
Do not discharge product into the environment without control. The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components.

13. Disposal Considerations

Observe national and local legal requirements.
Residues should be disposed of in the same manner as the substance/product.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

14. Transport Information

Domestic transport:

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable

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UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
	Marine pollutant: no
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
Proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information**Other regulations**

16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.